

UI UX Case Study For Refy

smart livestock management platform

Refy is an AI-powered livestock management platform that helps farmers centralize animal records, monitor herd health, automate farm operations, and make smarter decisions through real-time insights and intelligent recommendations.

Tools Used



PROJECT OVERVIEW

The Project & The Problem & The Goals

Project overview

Reefy is an AI-powered livestock management platform designed to simplify farm operations. It enables farmers to monitor animal health, manage records, assign tasks, and detect diseases early through AI, helping improve productivity while reducing manual work.



The Problem

Farmers still rely on paper records and disconnected tools, making it difficult to track animal health, manage breeding schedules, organize daily tasks, and detect diseases before they become critical. This often leads to delayed decisions, higher costs, and productivity loss.

The Goals

Design a centralized, user-friendly platform that digitizes farm management, automates repetitive tasks, provides AI-driven health monitoring, and enables farmers to make faster and more informed decisions from one place.



DESIGN PROCESS

Design Thinking Methodology

Design Process

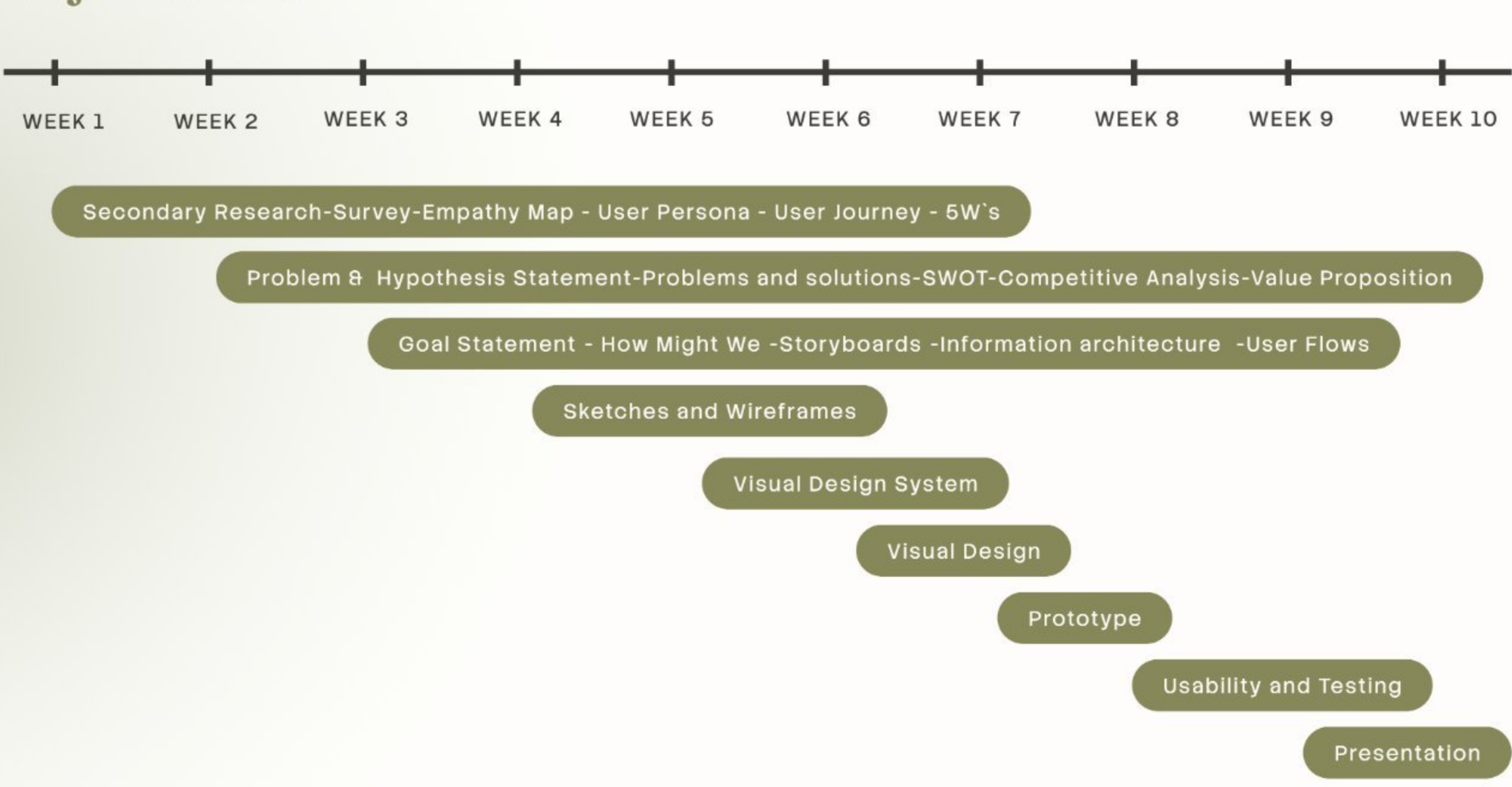
By using the Design Thinking methodology, I focused on understanding the needs of users and applied what I learned throughout the process to the project.



PROJECT TIMELINE

Timeline For Our Steps

Project Timeline



01. Empathize

Understanding Farmers' Needs Through Research & Insights

- Secondary Research
- Survey
- Empathy Map
- User Persona
- User Journey
- 5 W's

EMPATHIZE

Design Thinking Process - Secondary Research

Secondary Research

Description Before the design process, I conducted secondary research to understand the current state of livestock farming in Egypt, the challenges farmers face, and how technology can improve productivity and reduce economic losses.

Growing Livestock Population

Egypt has more than 18 million livestock animals, while most farms are still managed manually using paper records or WhatsApp notes, making data difficult to organize and retrieve.



High Economic Losses

Livestock diseases can reduce farm income by 20–30% annually due to delayed diagnosis, treatment costs, and animal mortality.

Poor Record Management

Most small and medium farms struggle to track vaccination schedules, breeding history, milk production, and medical records, leading to missed treatments and inaccurate decisions.



Low Digital Adoption

Although smartphones are widely available, digital farm management tools are still rarely used because existing systems are expensive, complex, and not designed for local farmers.

Delayed Disease Detection

Farmers often notice health issues too late. Early detection supported by AI and smart monitoring can significantly reduce disease spread and improve recovery rates.



Market & Financial Challenges

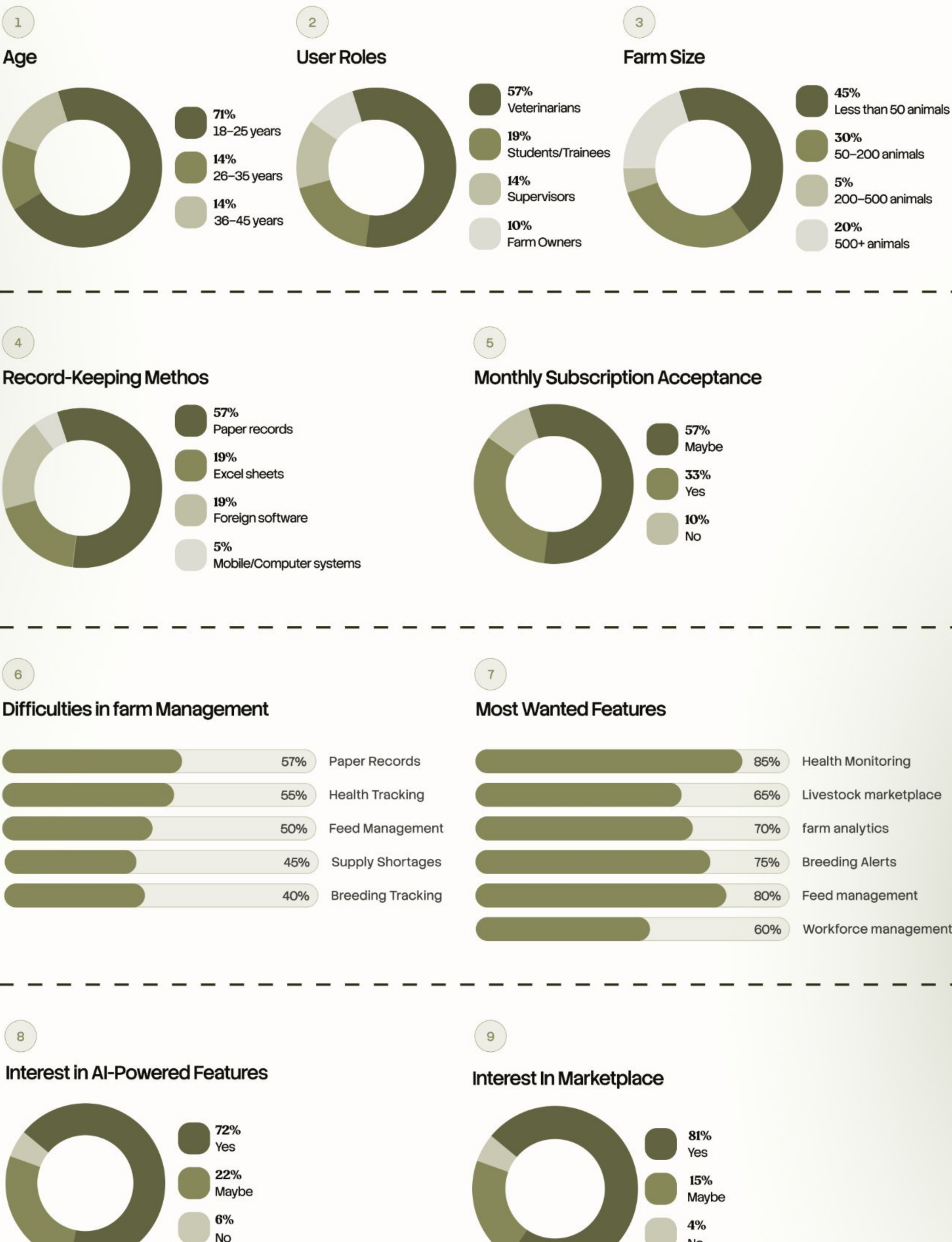
Farmers face unstable livestock prices, limited market visibility, and difficulty comparing suppliers, making buying and selling less efficient.

EMPATHIZE

Design Thinking Process - User Survey

User Survey

To measure users' preferences and behaviors, I prepared an online survey and created quantitative questions suitable for statistical analysis.



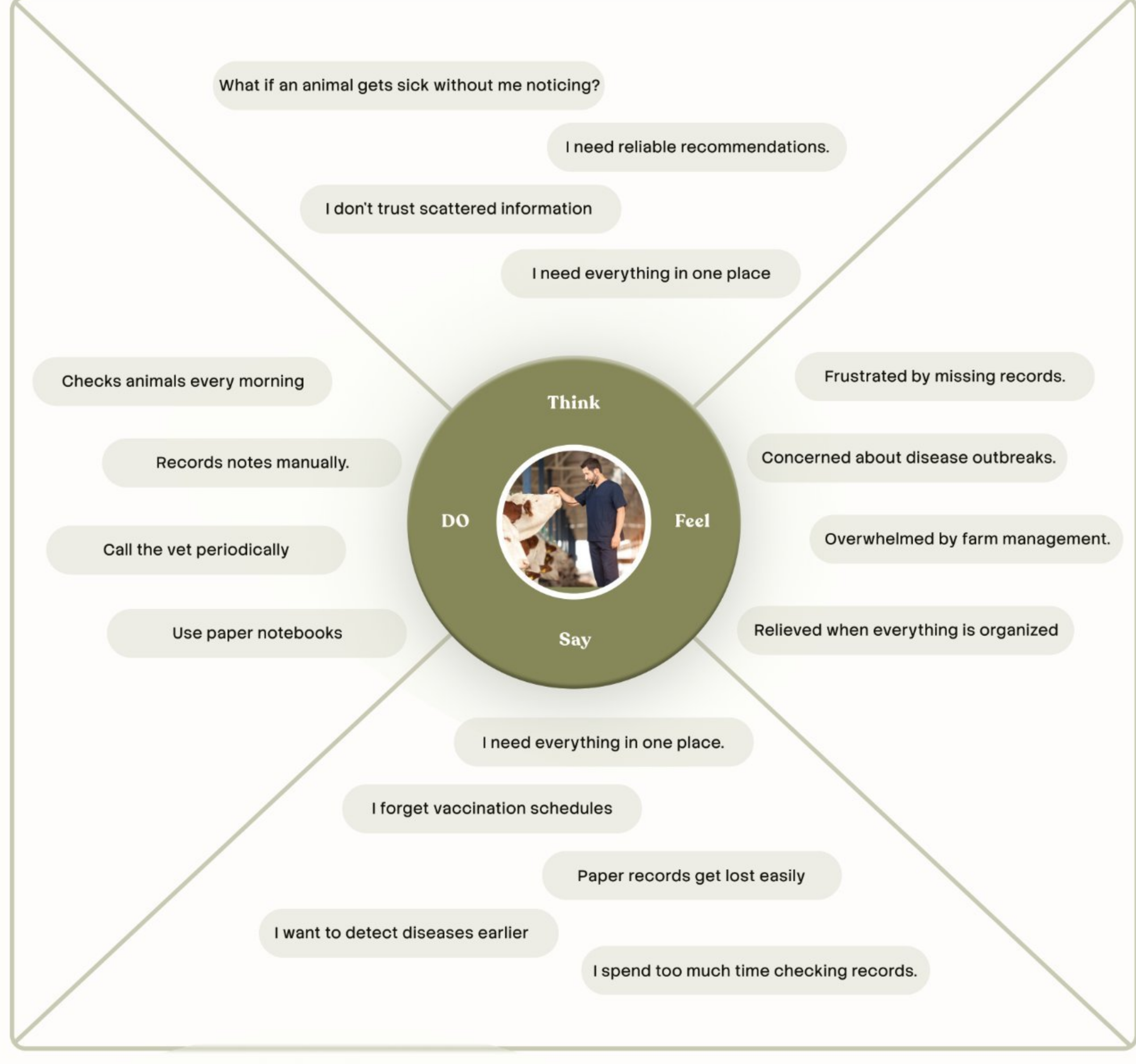
Survey Key Insights

- Paper-based management is still widely used.
- Animal health is the highest user priority.
- AI features received strong positive interest.
- Farmers value verified livestock marketplaces.
- Simplicity and ease of use drive adoption.
- Users are willing to pay for measurable value.
- Automation can reduce daily workload and farm losses.
- Mobile-first management fits farmers' daily workflow.



Empathy Map

By using Aggregated Empathy Maps, I consolidated multiple user perspectives, highlighted common patterns, needs, and pain points, and developed a more holistic understanding



User Personas

Fictional profiles representing our target users, built from real research insights.

Bio
Name : Ayman Ali
Age : 42 years old
Government : Menia

Farm Data
• Veterinarian at farm
• No of head: less than 50
• 1-3 worker
• Dairy and fattening farm

Statement
• Accounting profit and losses according to economy
• Easy entry of farm data
• Easy and readable daily of records
• Notifications must take a workflow of time (at first every 2 weeks then every week then every three days till become daily)
• Weighing for fattening section weekly

Needs
• Unified Dashboard: Fast data entry & rapid search in one screen.
• Offline & Auto-save: Fully functional without Wi-Fi.
• Smart Alerts: For feeding, calving, estrus, selling, and vaccines.
• Health & Milk: Herd health tracking and milk production reports.

Pain Points
• Complexity of returning to old data and information
• Long and difficult entry of data
• Reading daily records

App Usage
• Whatsapp as a note

Bio
Name : Ahmed Alghetany
Age : 24 Year
Government : Dakahlia

Farm Data
• Rancher
• No of head: less than 50
• 1-3 worker
• Dairy and fattening farm

Statement
• Reports and reminders for vaccination time one week before
• Reports for profit and losses
• Calving time reminder

Needs
• Smart System: Unified dashboard, fast entry/ search, offline auto-save, and multi-user access.
• Individual Tracking: Per-head profile (Age, Weight, Health), milk reports, and sickness alerts.
• Automated Reminders: For feeding, calving, estrus, selling, and vaccines (2 weeks prior).
• Financial Control: Automatic cost and profit tracking.

Pain Points
• Complexity of entry of data during farm labourer

App Usage
• Excel Sheet

Bio
Name : Michael Gerges
Age : 25 Year
Government : Giza

Farm Data
• Farm Supervisor
• No of head: 200-500
• 3-10 worker
• Fattening farm

Statement
• Recording unhealthy head in herd
• Reminders of vaccination time
• Tracking costs (purchasing bulls, fodder, medicine)

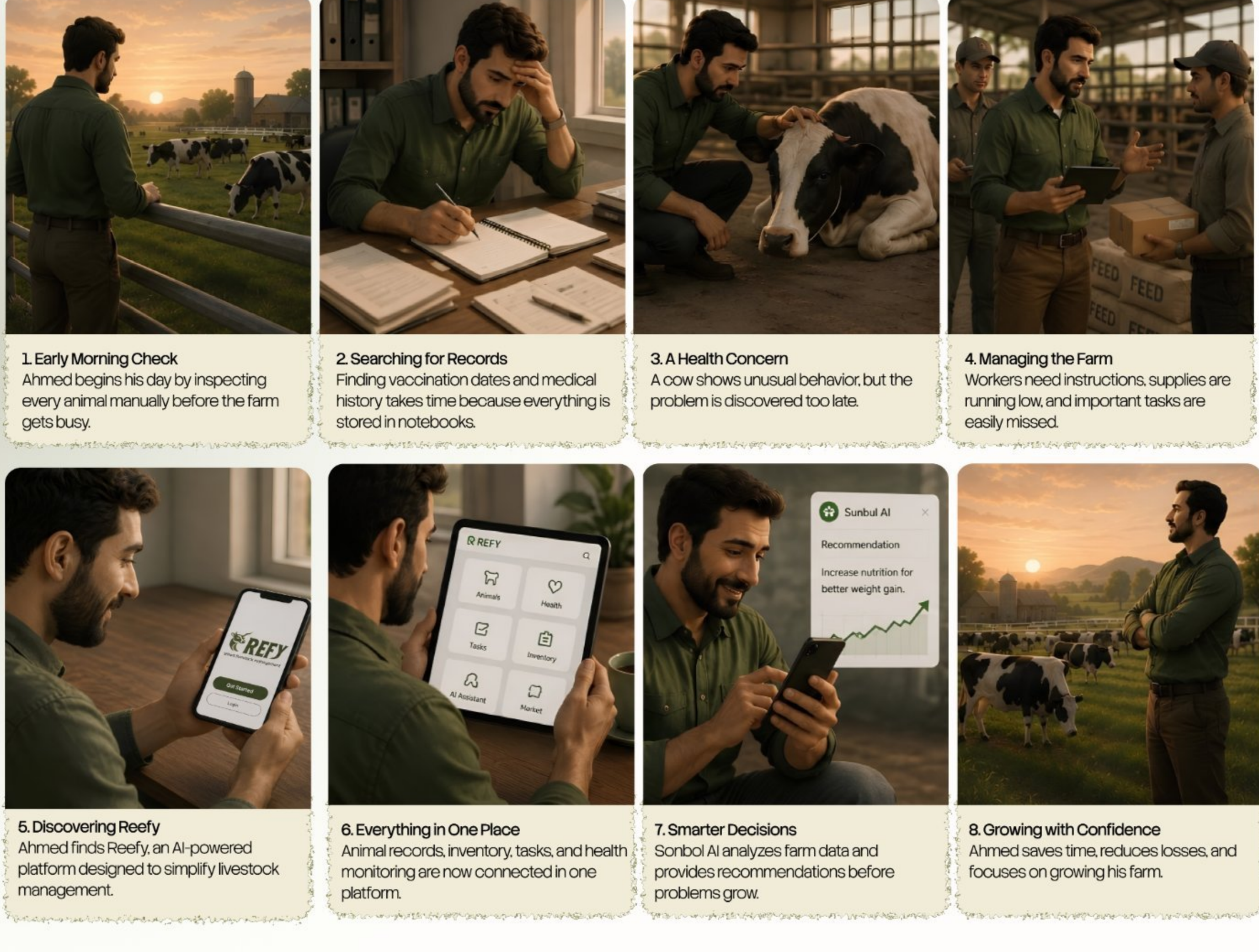
Needs
• Fast Dashboard: All-in-one screen with rapid data entry and search.
• Offline & Multi-User: Auto-saving without Wi-Fi and team access.
• Smart Alerts: Notifications for feeding, calving, selling, and monthly weighing.
• Health Tracking: Herd health monitoring with problem alerts.

Pain Points
• Missing medical records.
• Delayed diagnosis.
• Inconsistent communication with farmer

App Usage
• Dairy Cattle

User Journey

Mapping the User's Journey to Identify Pain Points & Opportunities



5W's Analysis

Clarifying who the product is for, what it solves, and why it matters.



02. Define

Synthesizing research findings into clear problems, opportunities, and strategic design decisions

- Problem & Hypothesis Statement
- Problems & Solutions
- Competitive Analysis
- SWOT Analysis
- Value Proposition

Problem & Hypothesis Statement

Problem Statement Clearly Define Users' Needs And Challenges. While Hypothesis Statements Predict Potential Solutions And Their Impacts. These Two Approaches Guide The Project In Developing User - Centered Solutions.



Problems & Solutions

From user pain points to meaningful design solutions.



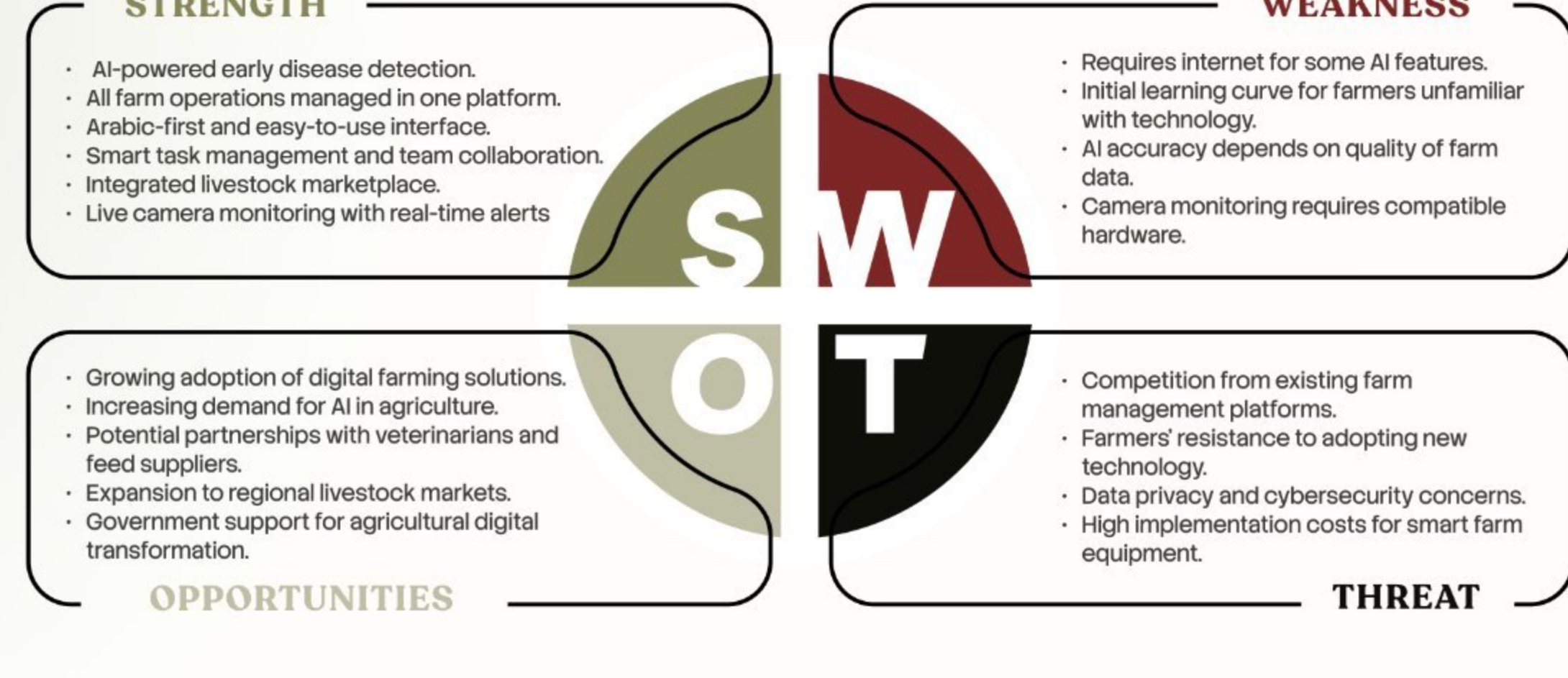
Competitive Analysis

Analyzing Existing Solutions To Identify Market Gaps And Opportunities.

Features	Cattly	cattlics	greener herd	Refy
Animal Management				
Health Records				
Inventory (feed, meds, hay)				
Daily Tasks				
Team Management				
Free				
AI Assistant				
AI Camera				
Marketplace				
Arabic Experience				

Swot Analysis

Evaluating Reefy's Internal Strengths And Weaknesses Alongside External Opportunities And Threats To Identify Strategic Advantages And Potential Challenges.



Value Proposition

The Unique Value Reefy Delivers By Solving Farmers' Daily Challenges Through A Smart, AI-Powered Farm Management Experience.



03. Ideate

Transforming research insights into practical concepts, structured user journeys, and solution-focused experiences.

- Goal Statement
- How Might We
- Storyboards
- Information architecture
- User Flows
- Sketches

Goal Statement

Our platform will let farmers manage livestock health, daily operations, and farm records from one place, which will affect small and medium farm owners by reducing manual work, improving decision-making, and enabling early disease detection.

We will measure effectiveness by:

- Reducing time spent on daily farm management.
- Increasing health record completion.
- Improving early disease detection.
- Increasing AI feature adoption.
- Improving task completion by farm workers.



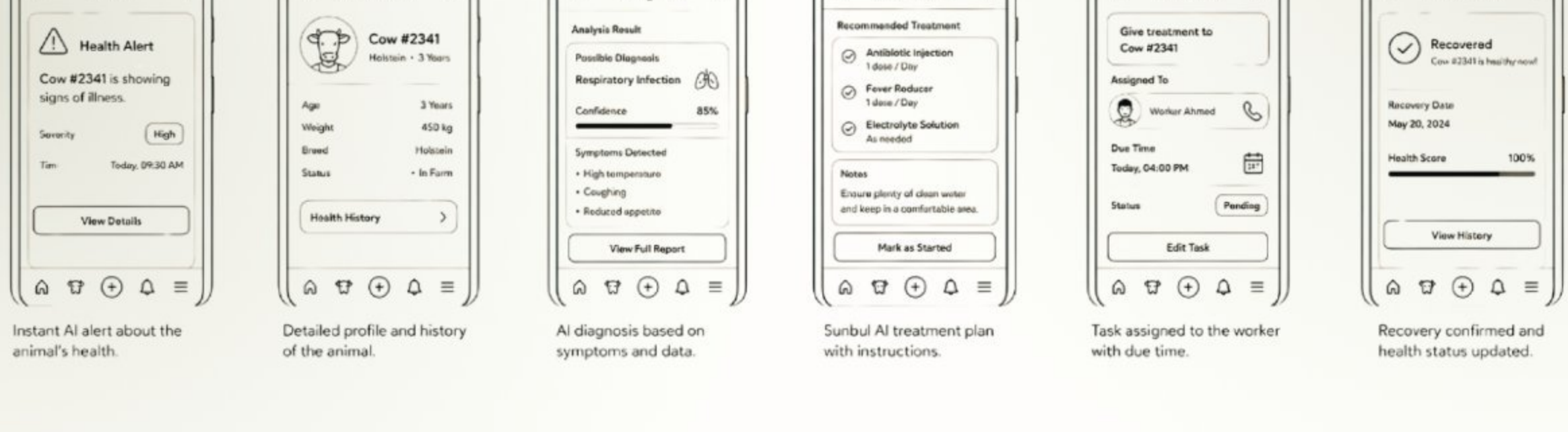
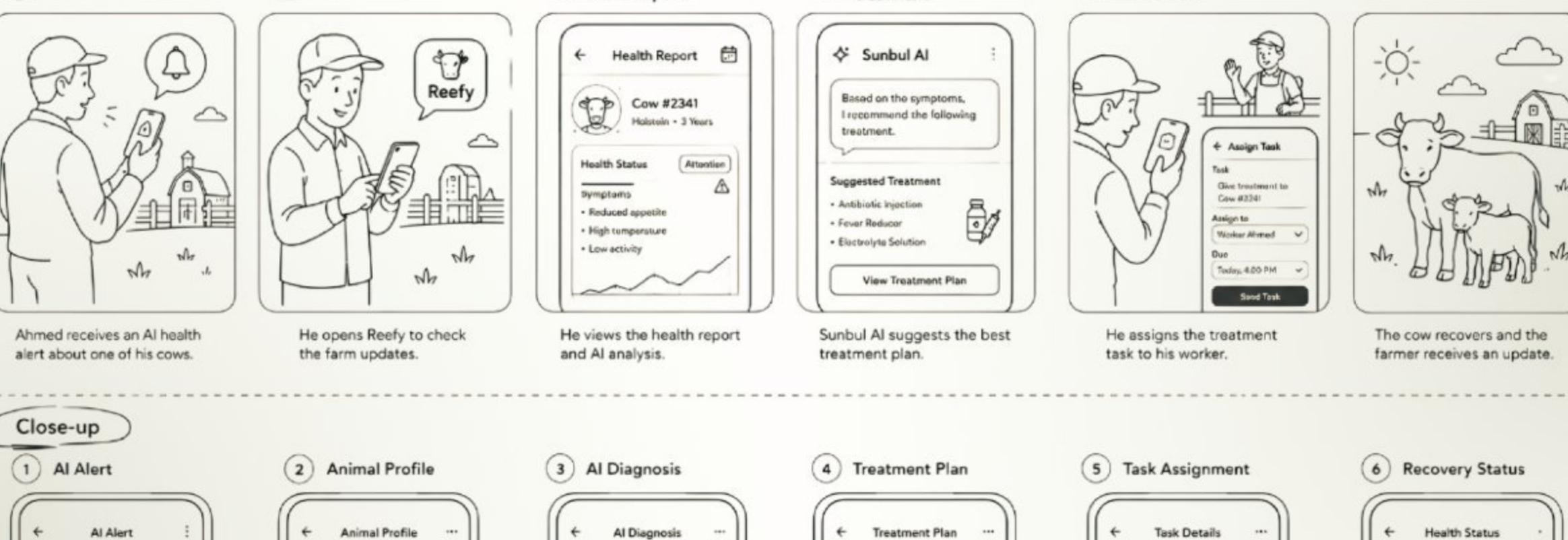
Brainstorm (How Might We)

To Generate Creative Solutions For Users' Needs And Challenges, I Crafted 'How Might We?' Questions. These Questions Guide The Project In Developing Innovative And User-Centered Ideas

- How Might We Detect Diseases Before Symptoms Become Severe?
- How Might We Simplify Daily Farm Management?
- How Might We Reduce Manual Paperwork?
- How Might We Help Farmers Track Every Animal Easily?
- How Might We Improve Communication With Farm Workers?
- How Might We Make Livestock Trading More Trustworthy?
- How Might We Make AI Easy For Every Farmer?
- How Might We Help Farmers Make Faster Decisions?

Storyboards

Storyboards Help Visualize The User Journey By Showing How The Product Fits Into Real-Life Scenarios. Both Big Picture And Close-Up Perspectives Are Used To Highlight User Needs, Actions, And Emotions Throughout The Experience



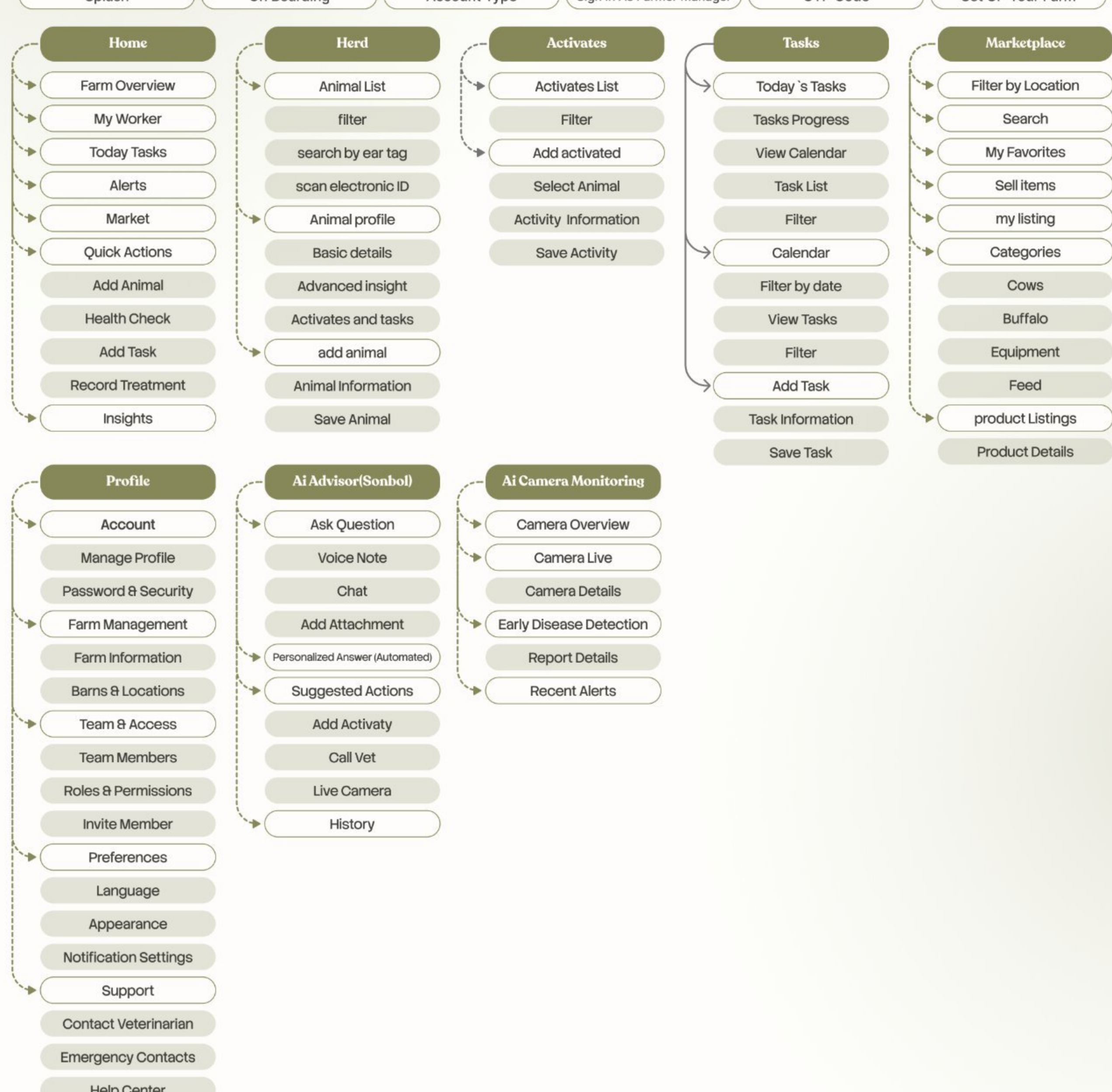
Information Architecture

Information Architecture (IA) Organizes Content So Users Can Easily Navigate And Find What They Need. A Site Map Visualize This Structure, Showing How Pages Are Features Are Connected

Website IA



App IA



Q: Can I use the platform offline?

Q: Can I use the platform offline?

Great question. We understand that reliable internet connectivity can sometimes be a challenge on farms, especially in rural areas.

Offline Functionality:

Limited Offline Mode: The platform has limited offline functionality for critical daily tasks. If your internet connection drops, you can still:

- Record basic animal information (weight, health observations, feed consumption)

- Add notes and daily activity logs

- Record basic operational tasks

These records are saved locally on your device (phone, tablet, or computer) and automatically sync to our servers once your internet connection is restored. This means you won't lose any data, even if your connection is intermittent.

Important Notes: Offline mode is limited to essential record-keeping tasks. More complex features like the AI Smart Advisor, market listings, and some advanced reporting require an internet connection.

How Syncing Works:

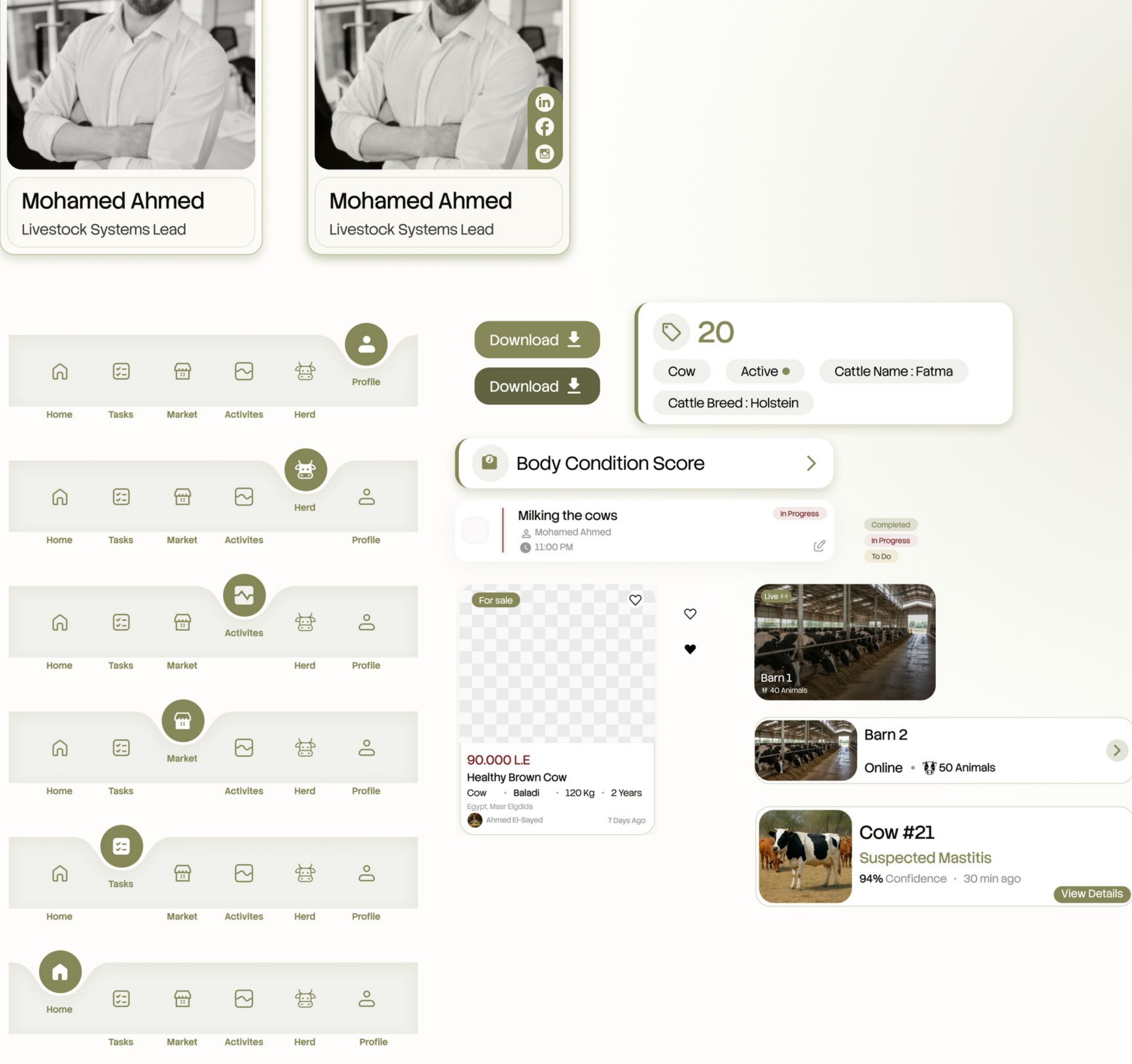
When you're offline, the app works with a local copy of your farm data. As soon as your internet connection returns, any changes you made offline are automatically synced to the server. This syncing happens in the background without you needing to do anything.

If you and another team member both make changes to the same animal record while offline, our system detects this when you reconnect and shows you both versions so you can decide which information to keep. This ensures data consistency.

Internet Reliability:

If you regularly experience internet issues, we recommend:

1. Checking your mobile data signal in your farm area
2. Considering a wireless router or mobile hotspot for more stable connectivity
3. Downloading relevant help articles before going offline for extended periods



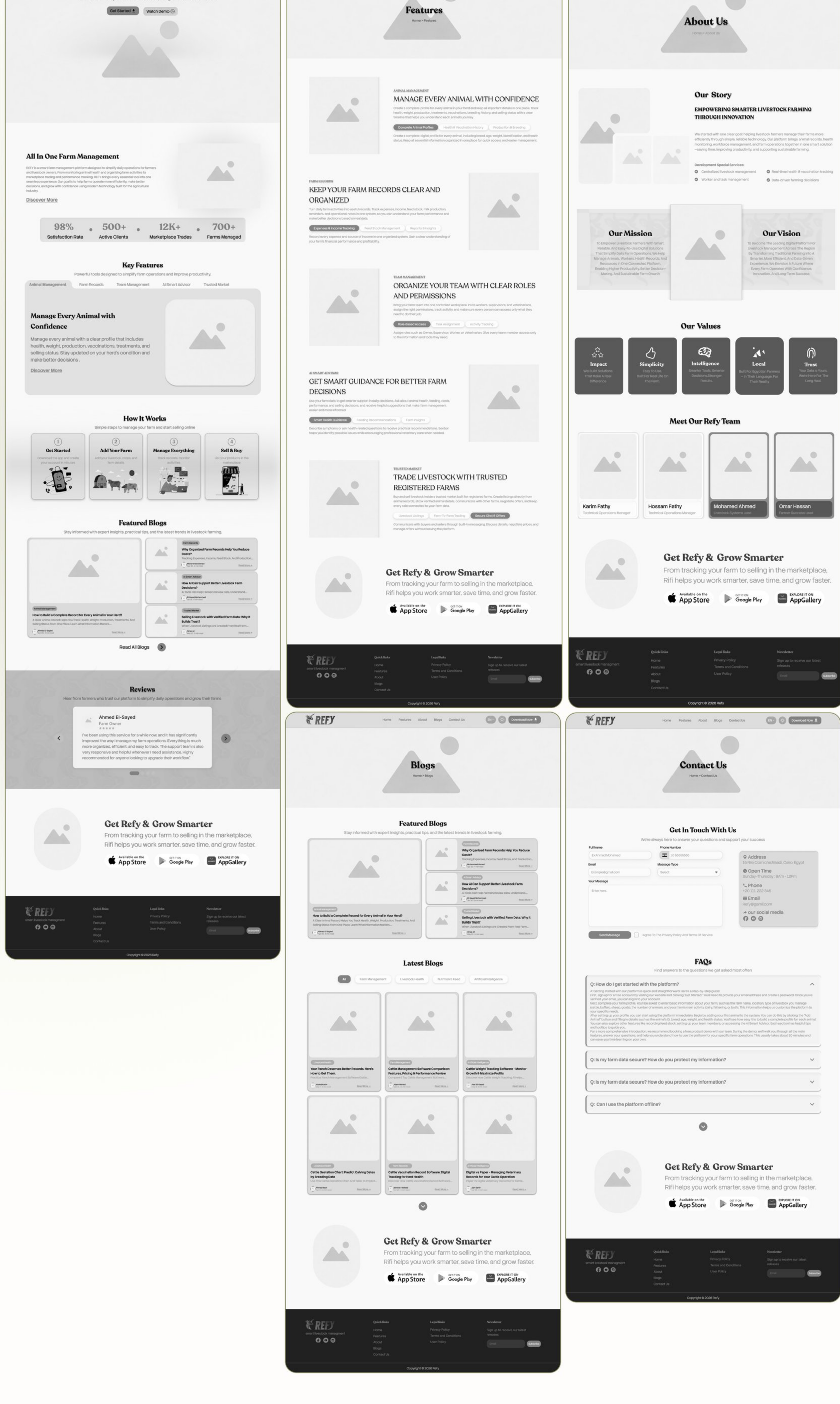
DESIGN

Design Thinking Process - Wireframe

Medium Fidelity Wireframe

Information Architecture (IA) Organizes Content So Users Can Easily Navigate And Find What They Need. A Site Map Visualize This Structure, Showing How Pages Are Features Are Connected

Website



Application



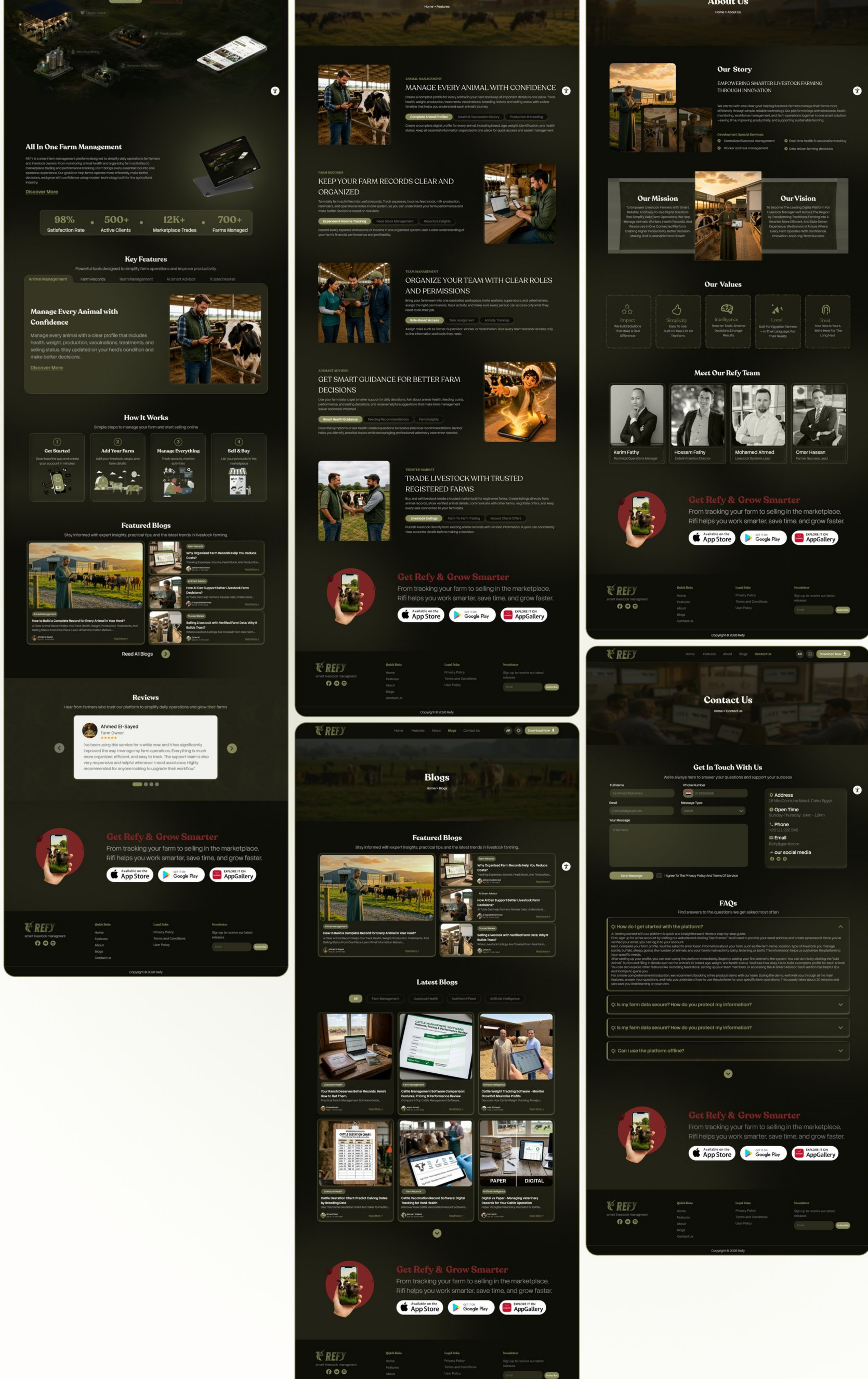
DESIGN

Design Thinking Process - Visual Design

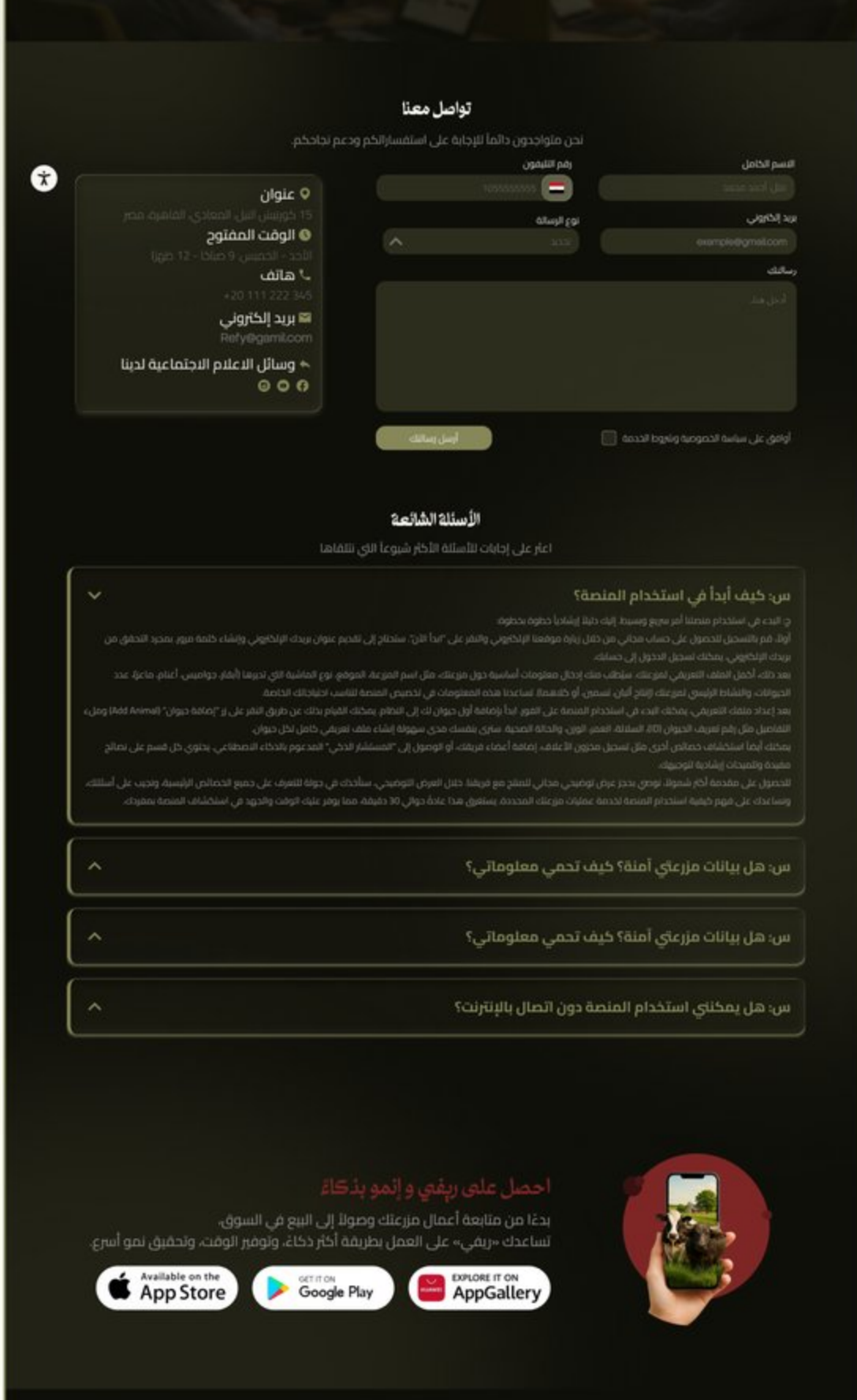
Visual Design

Transforming Wireframes Into A Modern, Intuitive, And Consistent Interface Tailored For Livestock Farm Management.

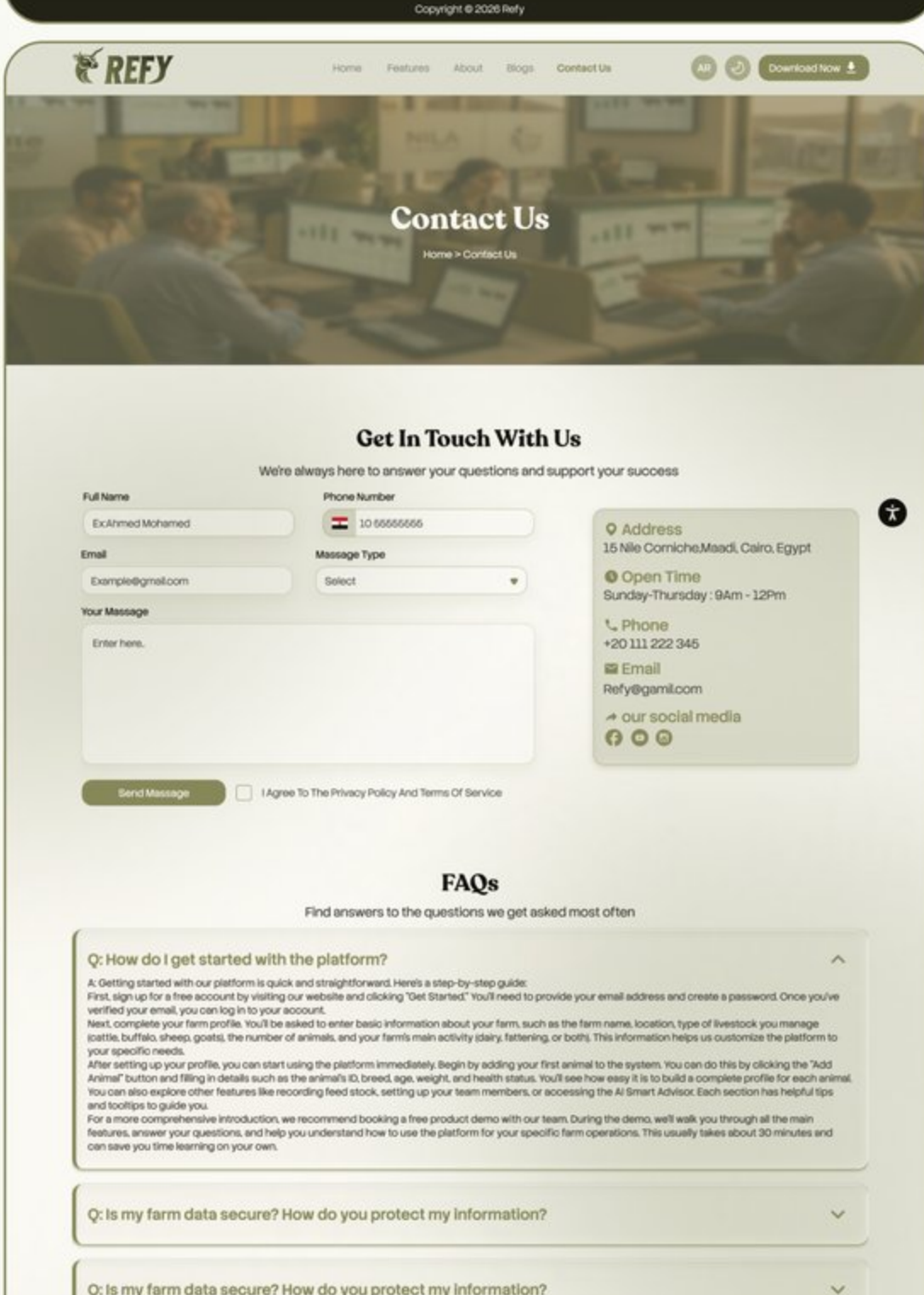
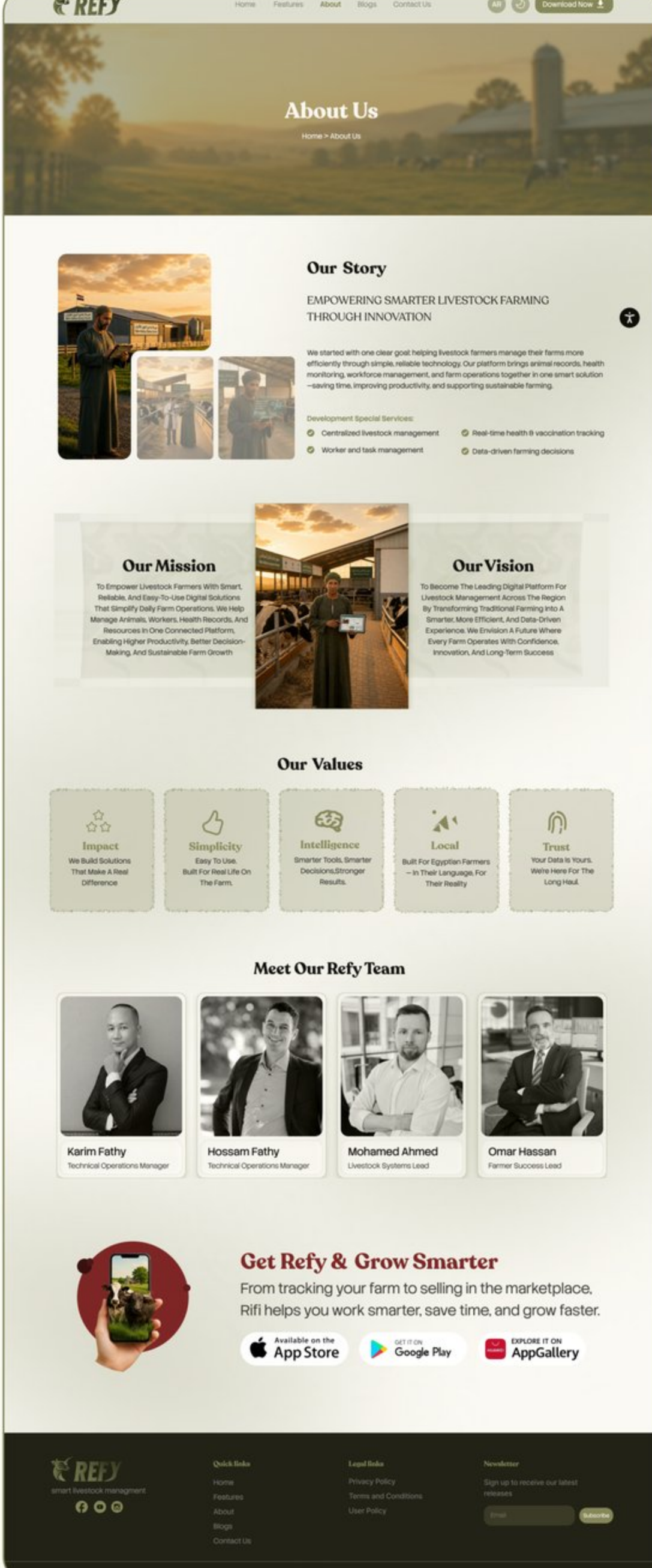
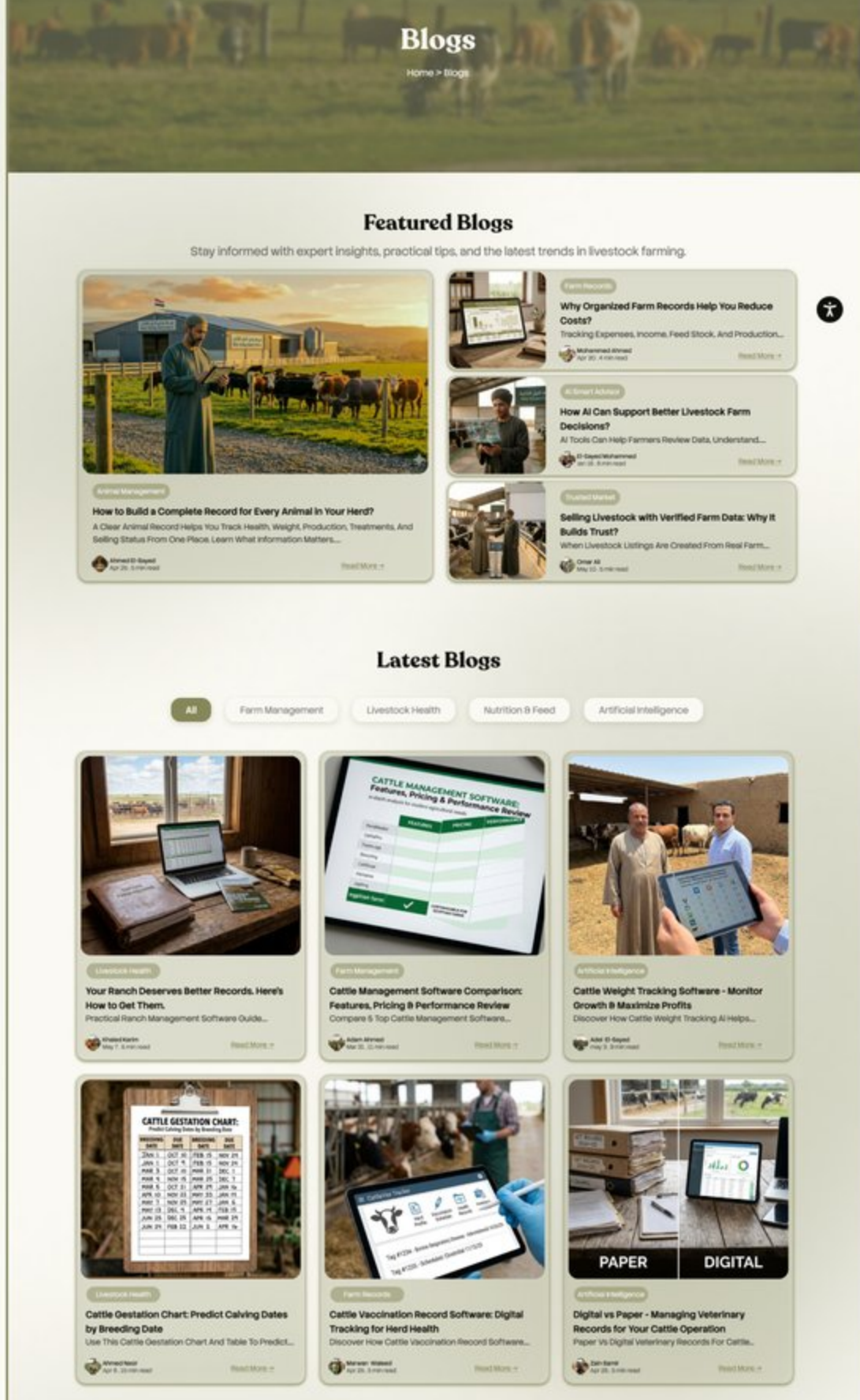
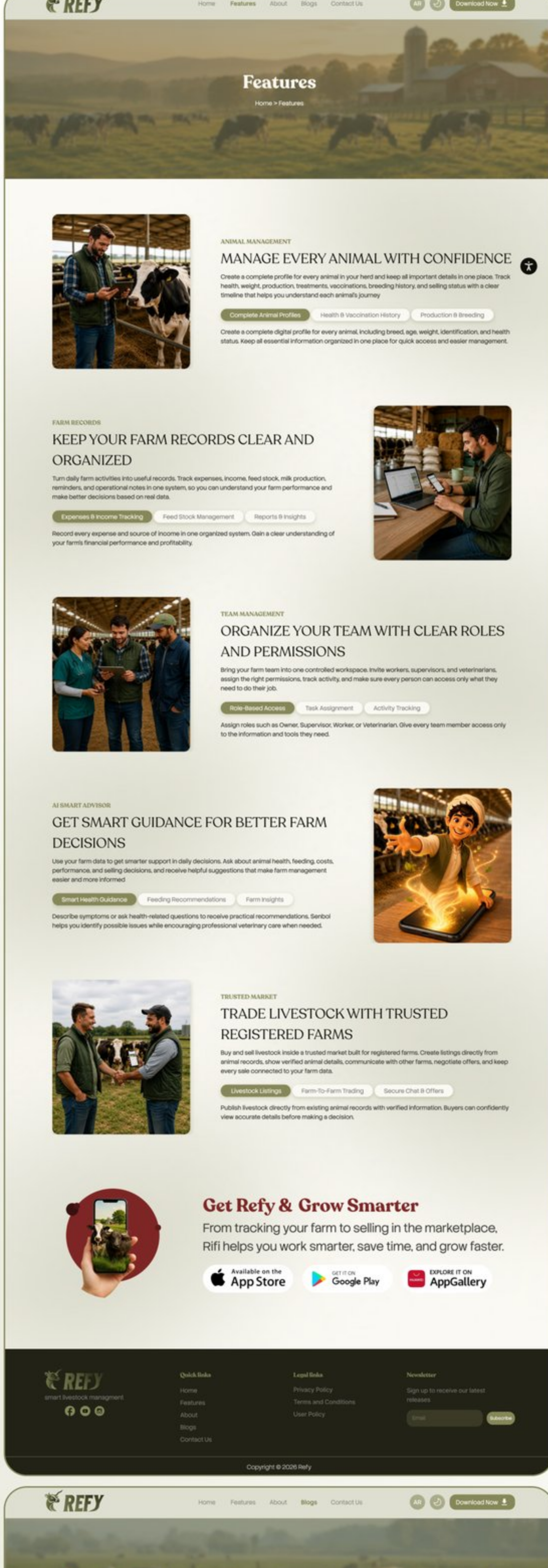
Website - Dark Mode - English



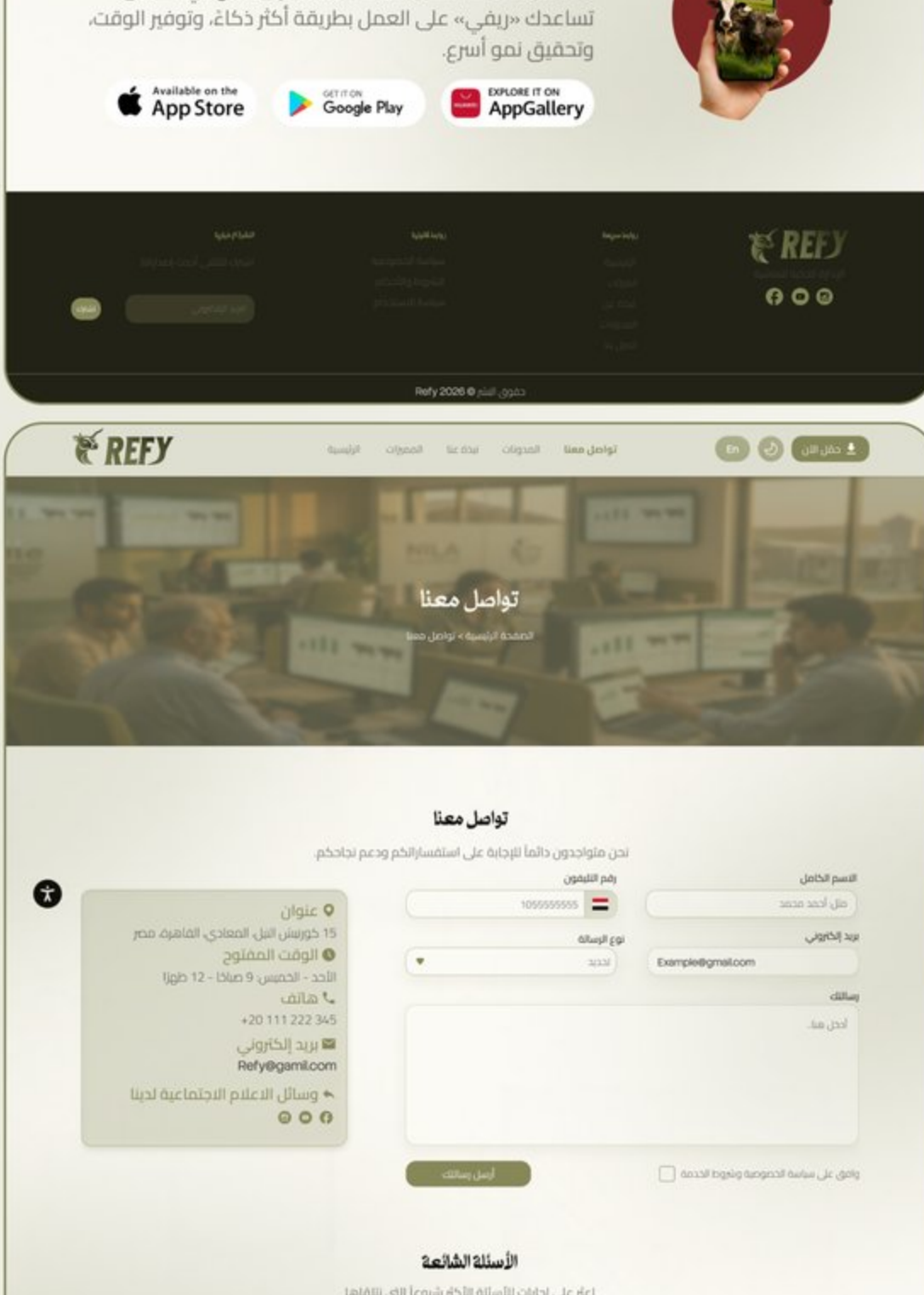
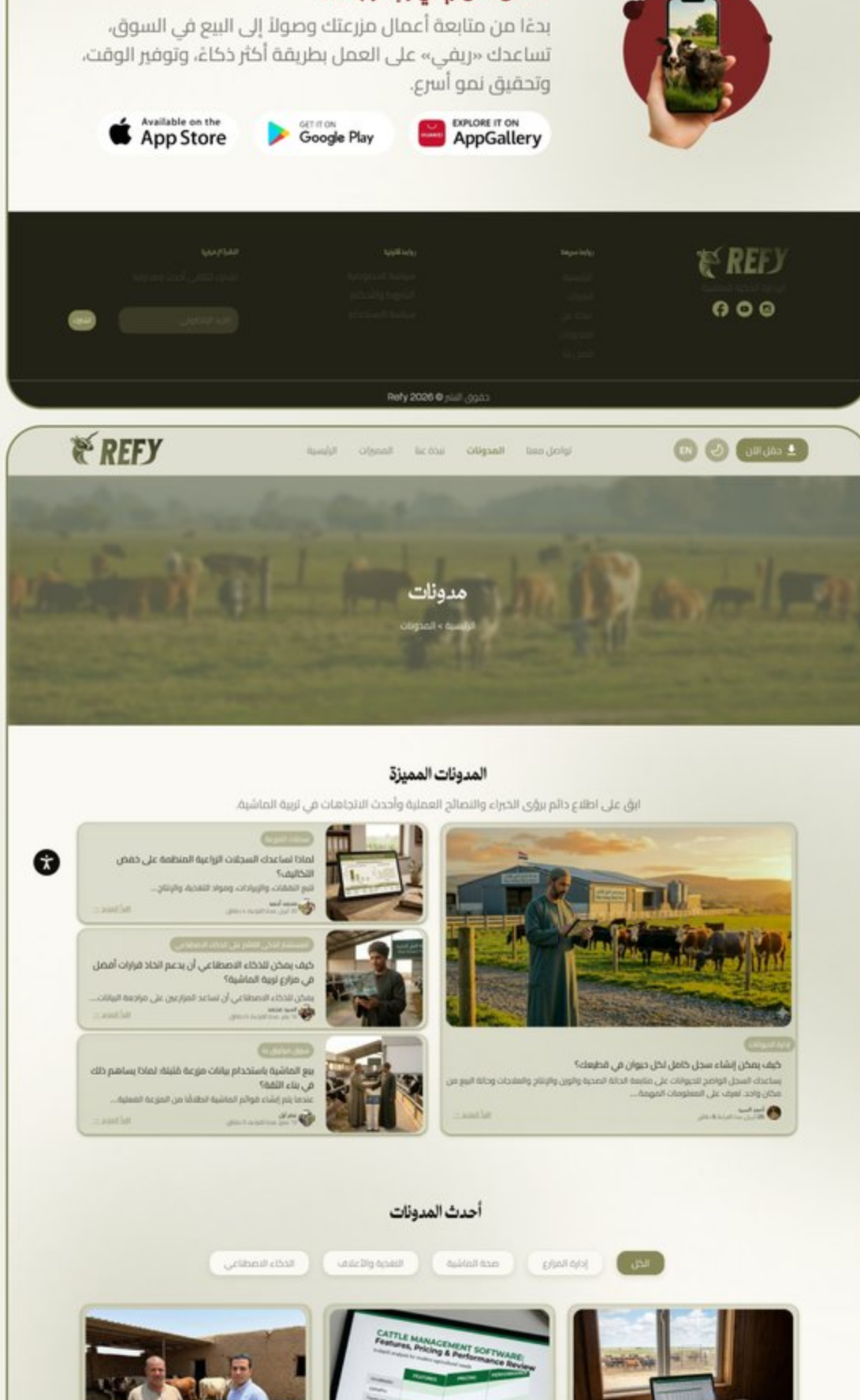
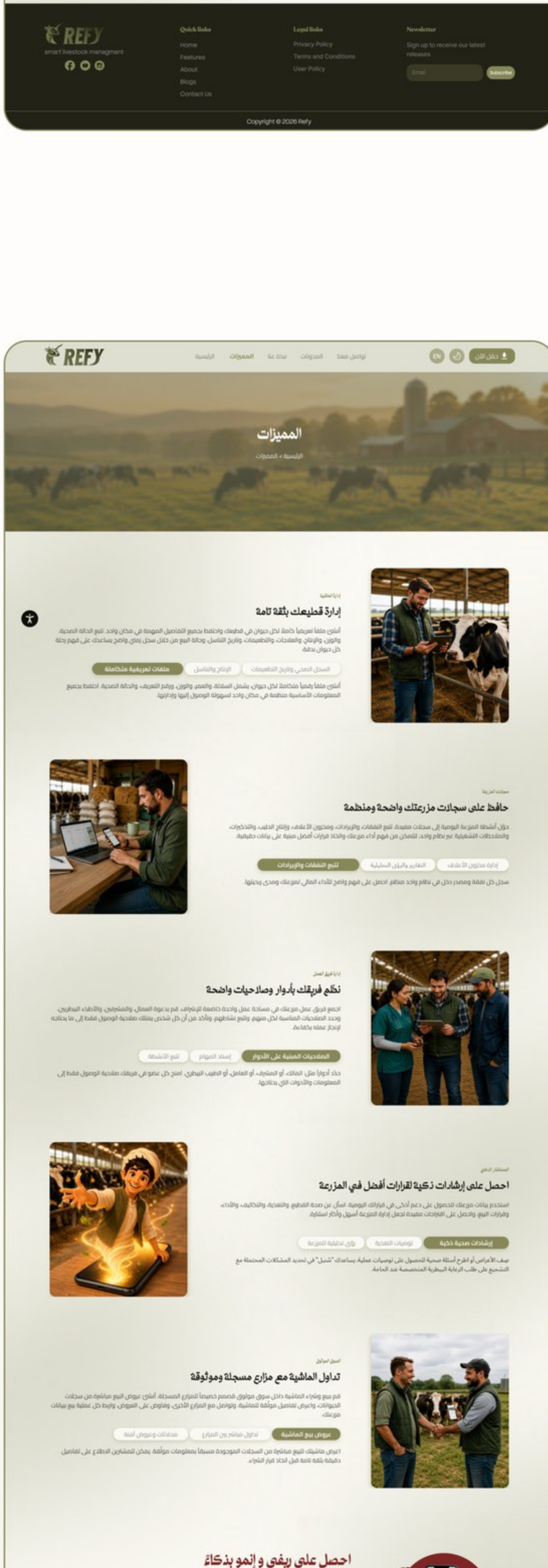
Website - Dark Mode - Arabic



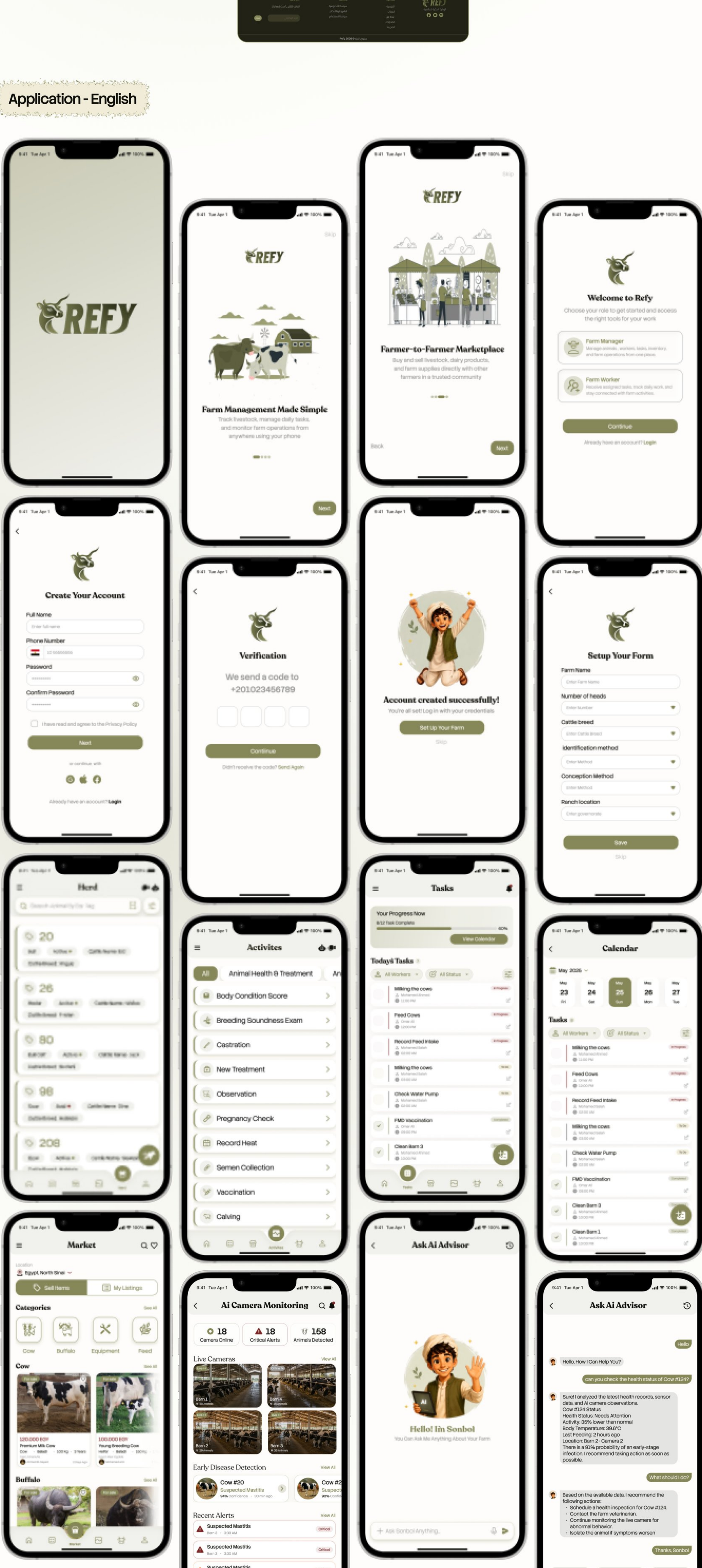
Website - Light Mode - English



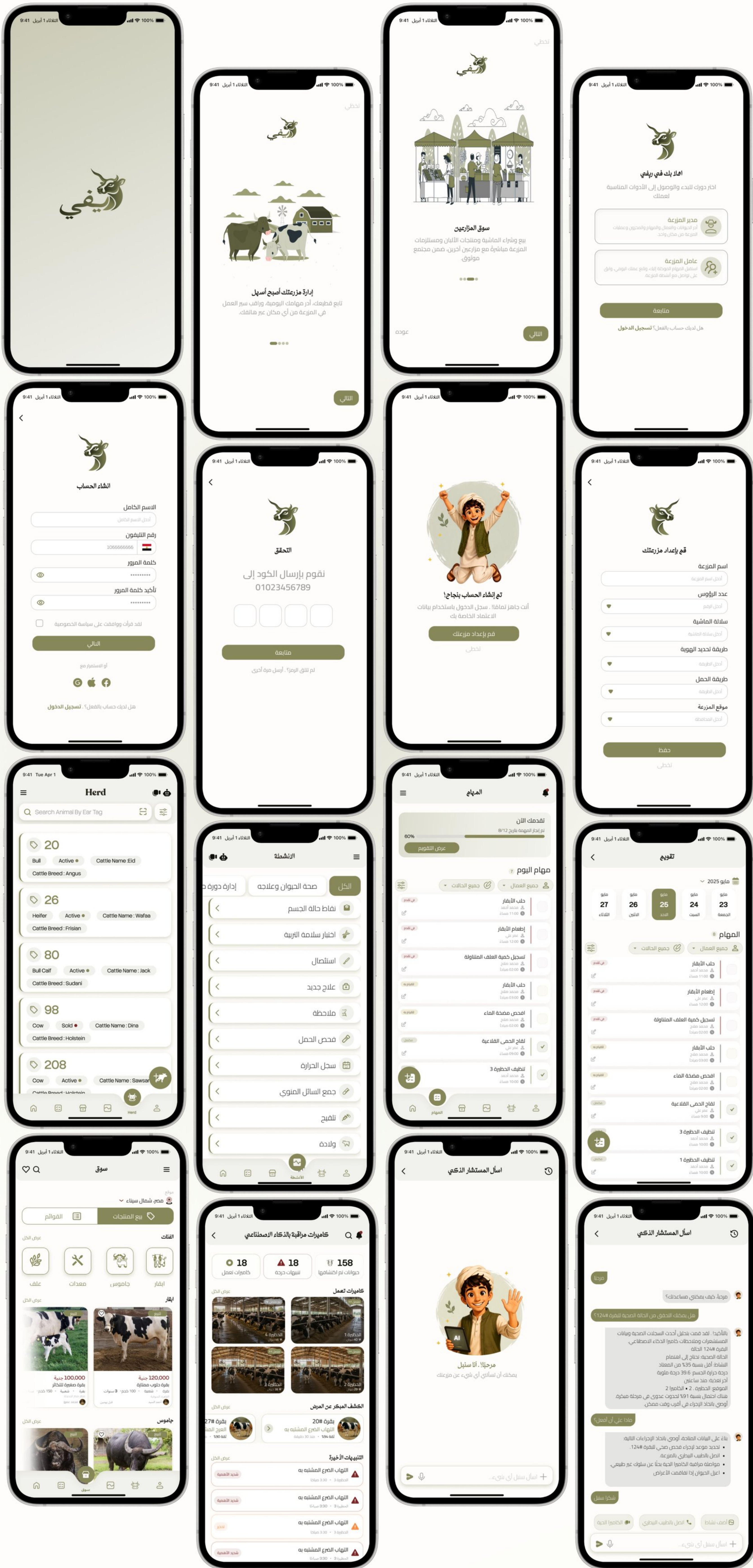
Website - Light Mode - Arabic



Application - English



Application - Arabic



“

Take care of your cows, and they will take care of you

Temple Grandin

”

Thank You

